

Future Sales Prediction using Machine Learning and Deep Learning Techniques



Dhanesh Rodriguse (ID 10610107)

Supervisor : Prof. Amit Sharma

Dublin Business School

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Declaration

I hereby declare that the thesis entitled "*Future Sales Prediction using Machine Learning and Deep Learning Techniques*", submitted for the completion of *Master of Science in Business Analytics* at *Dublin Business School*, is entirely my original work. I affirm that this work has not been submitted, in part or in whole, for the award of any other academic degree or qualification in any other institution or university.

I further declare that all sources of information used or referred to in this thesis have been duly acknowledged and cited according to the established academic conventions. Any materials or ideas obtained from external sources have been clearly identified, and their usage is accompanied by appropriate references and attributions.

Dhanesh Rodriguse (ID 10610107)

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Abstract

This study investigates advanced machine learning and deep learning techniques for accurate sales forecasting using the Amazon Sales Dataset. By analyzing diverse attributes including date, store, department, and economic indicators, our study aims to develop robust prediction models. Techniques such as attention mechanisms, graph neural networks (GNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, autoregressive integrated moving average (ARIMA), and vector autoregression (VAR) models are explored. Experimental results highlight the strengths of deep learning in capturing complex patterns, enhancing predictive accuracy over traditional models. We showcase the role of each technique in different scenarios, considering interpretability and performance. Ensemble methods are also examined to further boost forecasting accuracy. Our research contributes to effective sales prediction, empowering businesses with informed decision-making for optimized resource allocation and operational efficiency.

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Chapter 1

1. Introduction

Accurate sales prediction plays a pivotal role in modern business strategies, enabling effective resource allocation, inventory management, and decision-making. This thesis embarks on a comprehensive exploration of sales forecasting, leveraging the capabilities of state-of-the-art machine learning and deep learning techniques. The focal point of this study is the Amazon Sales Dataset, a rich repository encompassing various dimensions such as date, store, department, and economic indicators. The objective is to develop advanced prediction models that can uncover intricate patterns and dependencies embedded within this multifaceted dataset.

In pursuit of this goal, we delve into an array of methodologies, including attention mechanisms, graph neural networks (GNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and classical time series models like autoregressive integrated moving average (ARIMA) and vector autoregression (VAR). Through rigorous empirical evaluations and comparative analyses, we not only seek to ascertain the predictive performance of each technique but also to unravel their applicability in distinct scenarios.

By shedding light on the strengths and limitations of these methods, we provide valuable insights into the landscape of sales forecasting. The ultimate aim is to equip businesses with versatile tools that facilitate precise predictions, thereby fostering informed decision-making, optimized operations, and enhanced competitiveness.

1.1 Back Ground scope

Sales forecasting has emerged as a critical component in modern business operations, influencing decision-making, resource allocation, and overall strategic planning. Accurate predictions empower organizations to optimize inventory levels, manage supply chains effectively, and align marketing strategies with anticipated demand. With the advent of advanced machine learning and deep learning techniques, the field of sales forecasting has witnessed a significant transformation, offering unprecedented insights into complex consumer behaviours and market trends.

Historically, sales forecasting has relied on traditional statistical methods such as time series analysis and linear regression (Box et al., 2015). However, these methods often struggle to capture the nonlinear patterns, dependencies, and seasonality inherent in sales data, especially in the presence of intricate interrelationships between various factors. This limitation has prompted the exploration of more sophisticated techniques that can better handle the dynamic and multidimensional nature of sales data.

Machine learning (ML) and deep learning (DL) have emerged as formidable tools for tackling the challenges posed by sales forecasting. ML algorithms, such as decision trees, random forests, and support vector machines, offer the ability to capture complex patterns in data without explicit programming. On the other hand, DL techniques, including recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, excel in sequential data analysis, making them well-suited for time-dependent sales data.

In recent years, attention mechanisms and graph neural networks (GNNs) have gained prominence for their capability to handle intricate relationships within data. Attention mechanisms allow the model to focus on specific elements of the input, effectively learning to weigh the importance of different attributes. GNNs, which extend neural networks to graph-structured data, are particularly valuable for sales forecasting when considering the interconnectedness of factors like store attributes, economic indicators, and consumer behaviour.

In addition to these modern techniques, traditional time series models like autoregressive integrated moving average (ARIMA) and vector autoregression (VAR) continue to be relevant benchmarks in sales forecasting. These models provide a foundation for understanding temporal patterns and serve as a baseline for evaluating the predictive performance of more complex approaches.

The Amazon Sales Dataset, chosen for this research, offers a rich and diverse environment to explore these advanced techniques. This dataset not only includes standard sales and economic indicators but also encompasses categorical variables like store type and department. Such complexities make it an ideal candidate for testing the efficacy of various methods in capturing intricate relationships and patterns.

Furthermore, the utilization of ensemble techniques presents an opportunity to harness the strengths of multiple models. Ensembling, achieved through methods like stacking or boosting, can enhance predictive accuracy by combining the outputs of different models, thereby mitigating biases and errors inherent in individual models.

In summary, this thesis aims to bridge the gap between traditional and modern sales forecasting techniques by exploring the potential of advanced machine learning and deep learning methods. By harnessing the power of attention mechanisms, GNNs, RNNs, LSTMs, ARIMA, and VAR models, this research seeks to contribute insights into their applicability and performance in the context of the Amazon Sales Dataset, ultimately providing businesses with accurate sales predictions for improved decision-making and operational efficiency.

1.2 Research Questions

The main research question of this study is:

1. How do advanced machine learning techniques, including attention mechanisms, GNNs, RNNs, LSTM, ARIMA, and VAR, compare in terms of accuracy and performance for sales forecasting using the "Predict Future Sales" dataset?

1.3 Objective

This Study aims to enhance sales prediction accuracy through a comprehensive exploration of advanced machine learning and deep learning techniques applied to the Amazon Sales Dataset. The primary objective is to develop and evaluate models incorporating attention mechanisms, graph neural networks (GNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, as well as traditional time series models like autoregressive integrated moving average (ARIMA) and vector autoregression (VAR). By comparing the effectiveness of these methods in capturing complex patterns and dependencies, this research seeks to provide businesses with refined tools for informed decision-making and optimized resource allocation in the realm of sales forecasting.

Chapter 2

2. Literature Review

2.1 Introduction

This section of the research paper will encompass an array of literature reviews focusing on existing relevant works pertaining to the research paper's subject matter, which is centered around the utilization of "machine learning" and "deep learning" techniques for predicting future sales. The primary objective of this chapter is to enhance comprehension of the research paper's theme. To facilitate a deeper understanding of the subject, it is imperative to initially grasp the keywords integral to the study's focus. On the basis of the topic, the keywords that can be identified are "Future Sales", "Future Sales Prediction", "machine learning" and "Deep Learning".

The estimation of future sales revenue and the forecasting of which deals will advance through the sales cycle are both referred to as "future sales," also known as sales forecasts (Pitani and Lohiya, 2022). Sales projections have an impact on immediate expenditure choices and important transaction choices. The process of "future sales prediction" or sales forecasting entails estimating the volume of goods or services that a sales unit—which can be a single salesperson, a sales team, or an entire company—will sell over the course of the upcoming week, month, quarter, or year in order to estimate future income (Amir et al., 2023). Sales forecasting is advantageous to the entire firm. Finance uses estimates to set budgets for hiring new personnel and planning capacity, and production uses projections of sales to arrange its cycles. Forecasts help sales strategy establish partner and channel strategies, the supply chain plan material purchases and production capacity, and sales operations plan territory and quotas.

Artificial intelligence (AI) and computer science's "machine learning" branch is concerned with employing data and algorithms to simulate human learning processes and gradually improve accuracy (Koçak et al., 2023). Integral to the evolving realm of data science, "machine learning" plays a pivotal role. This involves employing statistical methodologies to instruct algorithms in generating classifications, predictions, and valuable insights during data mining endeavors. These insights subsequently influence decision-making processes across applications and enterprises, potentially bolstering critical growth key performance indicators. On a parallel note,

"Deep learning" constitutes a subset of "machine learning," encompassing neural networks comprising three or more layers, as elucidated by Sarker (2021). These neural networks strive to replicate the intricate functioning of the human brain, enabling them to 'learn' from vast datasets, albeit without matching its cognitive prowess. Although unable to replicate its capabilities, neural networks with multiple hidden layers can enhance and refine accuracy; even a single-layer network can produce approximations.

2.2 Concept of Future Sales Prediction Using ML and DL Techniques

Business forecasting and decision-making are being revolutionised by the idea of future sales prediction utilising "machine learning" and "Deep learning" approaches (Kunduru, 2023). These methods offer precise forecasts based on previous sales data, market trends, consumer behaviour, and other pertinent aspects by utilising the power of complex algorithms and data analysis. Some of the key components of the concert are discussed below.

Data Preparation: The process of gathering and preparing data is the initial stage in forecasting future sales (Wisesa *et al.*, 2020). This entails preparing the data for analysis, addressing missing values, and cleansing the data. It is possible to include historical sales data along with other pertinent information including marketing efforts, seasonality, economic factors, and customer profiles.

Feature Engineering: In the development of models, feature engineering is crucial. To train the "machine learning" or "Deep learning" models, suitable features must be chosen and created from the data that is already accessible (Mohammed *et al.*, 2021). Creating lagged variables, extracting time-based features, or developing new features that capture crucial facets of sales behaviour are all possible steps in this process.

Model Selection: For precise forecasts, selecting the right model is essential. There are several different machine-learning methods that can be used, including gradient boosting, support vector machines, decision trees, and random forests (Rong *et al.*, 2020). "Deep learning" methods such as recurrent neural networks (RNNs) or long short-term memory (LSTM) networks are frequently employed for time series analysis and capturing temporal dependencies.

Model Training and Evaluation: The chosen model is then trained using the pre-prepared data. To assess model performance, the dataset is often split into training and validation sets (Sharma, 2021). In order to reduce errors and increase accuracy, the model is trained using past sales data

and internal parameter adjustments. Performance can be improved by modifying the model's architecture and hyperparameters.

Forecasting and Visualization: Future sales estimates can be produced using the learned model. The model is given data from future time periods, and it produces sales estimates based on the discovered patterns and linkages. These forecasts offer insightful information about potential sales patterns and can help with decision-making. The anticipated sales patterns can be presented and interpreted more easily by using visualisation techniques like line graphs or heatmaps.

Monitoring and Refinement: For the purpose of evaluating the model's accuracy and making the required corrections, sales forecasts should be routinely compared to actual sales data (Behera and Nain, 2019). The model can be updated and retrained to take into account the most recent data as it becomes available. The model's accuracy and adaptability to shifting market dynamics are maintained by this iterative procedure.

The incorporation of "machine learning" and "Deep learning" for the purpose of predicting future sales provides a lot of benefits to businesses.

a. Accurate and Data-Driven Insights:

These strategies can find intricate patterns and connections in sales data that conventional approaches might miss by utilising cutting-edge algorithms. This makes it possible for companies to make data-driven decisions and predict sales trends with accuracy.

b. Enhanced Decision-Making:

Businesses can use accurate sales forecasts to optimise resource allocation, inventory control, marketing plans, and overall business planning (Bandari, 2019.). This enables companies to make wise decisions, streamline processes, and boost profitability.

c. Improved Customer Experience:

Businesses can successfully meet client expectations by adjusting their offers and marketing tactics by understanding sales patterns and consumer behaviour. Improved client retention, loyalty, and satisfaction result from this.

d. Competitive Advantage:

Businesses might acquire a market advantage by correctly predicting upcoming sales trends. They are able to take proactive action in response to shifting market dynamics, take advantage of new opportunities, and outperform the competition.

2.3 Existing literature review

As per the study of Pallathadka *et al.*, (2023), Artificial intelligence is being used in the food business to reduce human error and increase resource efficiency, as will be discussed in this article. By utilising a range of fitting algorithms for sales projections, artificial intelligence and data science can increase productivity in lodgings, cafés, dining establishments, online meal delivery services, and restaurants while improving their quality. By encouraging a more transparent supply chain management system, AI has the ability to greatly improve packaging, extend the lifespan of products, combine items on the menu, and enhance food safety. Automated farming, intelligent farming, and drones are now necessary for the future of the agricultural sector due to AI and ML.

According to Brei, (2020), In recent years, there has been a gain in interest in the subject of neural networks for predicting. In order to predict the sales of furniture, a public dataset containing a retail store's sales history is inspected. Several forecasting prototypes are used to achieve this goal. Initially, the study employs conventional time-series forecasting methods such as "Triple Exponential Smoothing" and "Seasonal Autoregressive Integrated Moving Average (SARIMA)." Subsequently, it delves into more advanced approaches encompassing "Long Short-Term Memory (LSTM)," "Prophet," and "Convolutional Neural Network (CNN)." Diverse methodologies for assessing precision are applied to juxtapose the performances of these models. The developments present the Stacked "LSTM" method's excellence over the alternative processes. The outcomes also show that the Prophet and CNN models performed well.

As per the discussion of Kumar *et al.*, (2021), the marketing industry has not yet experienced the extensive effects of artificial intelligence (AI) and "machine learning" (ML) in many societal sectors. Despite these flaws, ML offers a number of potential advantages, including the chance to use more reliable techniques for the extension of scientific discoveries. This monograph aims at reducing this shortage in four ways. First, to give marketing a comprehensive overview of "machine learning", which includes a look at its main subtypes and algorithms, application to marketing, and overall workflow. Second, to examine top-down and bottom-up learning approaches as two possible approaches for marketing researchers to learn ML. The third objective is to review recent working papers on a few interesting sub-fields of research in

marketing as well as "machine learning" (ML) applications that have been published in prestigious management and marketing journals, books, and book chapters. The monograph's final objective is to explore the effects of current trends and potential future breakthroughs in "machine learning" on the marketing industry.

Based on the study which has been conducted by Javed Awan *et al.*, (2021), Artificial intelligence is being used in the food business to reduce human error and increase resource efficiency, as will be discussed in this article. By utilising a range of fitting algorithms for sales projections, artificial intelligence and data science can increase productivity in lodgings, cafés, dining establishments, online meal delivery services, and restaurants while improving their quality. By encouraging a more transparent supply chain management system, AI has the ability to greatly improve packaging, extend the lifespan of products, integrate menu items, and improve food safety. Automated farming, intelligent farming, and drones are now necessary for the future of the agricultural sector due to AI and ML.

Drawing inspiration from Tao et al.'s investigation (2019), the researchers harnessed a "Black Friday sales dataset" sourced from Kaggle, featuring over 550,000 observations and scrutinizing 10 distinct qualitative and quantitative attributes within the domain of sales and purchases. Given the continuous nature of the predictive label, regression models emerged as fitting choices. Employing the MLlib library from the Apache Spark big data platform, two machine learning models—linear regression and random forest—were trained. These machine-learning techniques were employed to prognosticate forthcoming prices and sales.

Initially, in the absence of the Spark framework, a linear regression model and a random forest model were established, yielding precision rates of 68% and 74% sequentially. Subsequently, the endeavor extended to incorporate the "Spark machine learning" framework and the broader big data infrastructure. This adaptation resulted in improved outcomes, where the linear regression model showcased an accuracy of 72%, while the random forest model demonstrated an accuracy of 81%.

On the basis of the study which has been conducted by Carta *et al.*, (2021), By putting forth a semi-supervised, "Deep learning"-enabled analytical pipeline, examine aspect-based sentiment dynamics of online reviews to foster digital collaborations. "Deep learning" techniques are used in this approach to represent and categorise text. Additionally, handle aspect and sentiment analysis together, building on earlier work that dealt with aspect extraction and sentiment identification separately. Additionally, by examining aspect-based sentiment in time series, this work offers a fresh way of looking at the dynamics of aspect-based feelings. The results of this study have major ramifications for consumers, product/service suppliers, and other online review stakeholders' digital cooperation.

As per the study conducted by Rouf *et al.*, (2021), The use of computer-aided stock trading techniques has increased in prominence in recent years, largely due to its effectiveness in processing historical data through "machine learning" to forecast future market behaviour. The most successful methods for tackling this job combine a variety of classifier decisions in order to forecast future stock prices. However, it has been demonstrated that using price information in a single supervised classifier produces subpar results, mostly because market history is insufficient as a predictor of future market behaviour. In this research, the authors provide a multi-layer and multi-ensemble stock trader to address this problem. begin by preprocessing the data using a large number of deep neural networks.

Artificial intelligence (AI) was used in the healthcare and agriculture sectors to improve disease prediction, continuous monitoring, crop production, improved operational efficiency, supply chain management efficiently, and decreased water waste. On the basis of the study conducted by Rashid *et al.*, (2020), these goals were achieved with the main goal of inventing standard, dependable product quality control processes. "Deep learning" and "machine learning" are two of the most widely used AI approaches. These models help businesses, individuals, and governmental organisations predict the future and gain knowledge from data. "machine learning" models are now being developed to handle the complex and varied types of data present in applications which is real-world. This article discusses the use of artificial intelligence and "machine learning" in social studies, management, healthcare, and agriculture.

The value of stocks has been steadily rising, and as a result, stock trading has expanded to become a significant area for investment for many investors from finance. Researchers are able to predict the market using unusual textual information from the internet thanks to advanced trading models. As per the study which has been conducted by Rasekhschaffe *et al.*, (2019) With the application of contemporary "machine learning" techniques" like text data analytics and ensemble approaches, prediction accuracy has greatly increased. Stock market forecasting and analysis continue to rank among the most challenging study areas due to the unpredictable, dynamic, and chaotic nature of the data. In order to understand the systematics of "machine learning"-based systems for forecasting stock markets, this study implements a broad framework. After being retrieved from online databases and libraries like Scopus and the ACM digital library, results from the years 2011 to 2021 were critically examined.

Based on the study discussed by Lim and Zohren, (2021), One of the widely used methods for assessing the dynamic states in brain pictures using various types of algorithms is "functional magnetic resonance imaging (fMRI)". However, due to the abundance of algorithms available in the literature, it is a significant problem for any researcher to select an appropriate deep or "machine learning" algorithm for processing fMRI data. Each researcher must invest some effort in learning about the numerous methods and algorithms that are applied to fMRI data. The current research on fMRI data using machine and "Deep learning" methods are reviewed methodically in this work. This review paper's main objectives are to: (a) identify the usage of "Deep learning" algorithms and "machine learning" in fMRI; (b) identify "Deep learning" and "machine learning" research trends for the performance of fMRI; and (c) assist new fMRI-based researchers in properly integrating their novel findings into the existing field of fMRI research.

As per the study of Punia *et al.*, (2020), In quantitative finance, "machine learning" is a contentious and important topic. Whether "machine learning" techniques can be useful investment tools is still a topic of intense discussion. Even though contextual, nonlinear, and delicate associations can be found using "machine learning" algorithms, overfitting makes it difficult to find signals in noisy historical data. And outline some of the fundamental ideas behind "machine learning" and give a straightforward illustration of how investors might employ

these methods to forecast the cross-section of stock returns while minimising the danger of overfitting.

Based on the study conducted by Al-Mobayed *et al.*, (2020), The diversity of time-series datasets across various fields has led to the development of numerous deep-learning architectures. The common encoder and decoder architectures used in both one-step-ahead and multi-horizon time-series forecasting are surveyed in this article, along with an explanation of how each model incorporates temporal information into its predictions. Then, also discuss recent improvements in "Deep learning" hybrid models, which incorporate both neural networks and well-researched statistical model components. Finally, discusses a few ways "Deep learning" might help decision support using time-series data.

As per the study of Akinosho *et al.*, (2020), the random forest (RF) networks and “long short-term memory (LSTM)” networks used in "Deep learning" are combined in this paper to present a unique forecasting technique. The suggested approach has an accuracy advantage over previous forecasting techniques since it can handle intricate temporal and regression correlations. On a multivariate dataset from a retailer, tested the novel methodology. And compare the new proposal's predicting performance to that of RF, multiple regression, “ARIMAX”, “LSTM”, and neural networks. The new proposition appears to be statistically much better, according to empirical data that is used to predict performance criteria to quantify bias, accuracy, and variation. Our technique also assigns a relative relevance ranking to the explanatory variables.

In this study conducted by Sengupta *et al.*, (2020), an Artificial Neural Network (ANN) model was employed to assist vehicle dealers in identifying the many features of cars, including their location, manufacturers, and classification into different categories including "buying," "maintenance," "doors, people, lug_boot, safety," and "overall." In order to predict car acceptance, ANN was utilised. The findings demonstrated that the “ANN model had a 99.12% accuracy rate in predicting car approval”. The rating of a car's acceptance is largely influenced by the safety element. The comparison study method may be applied to all other fields and is appropriate for the assessment of car acceptability forecasting.

The “resource planning, risk management”, and “logistical issues” that plague the construction industry are well-known to frequently lead to design flaws, project delivery delays, cost overruns, and legal battles. Based on the study conducted by, Tarallo *et al.*, (2019), to assist with the prescriptive and diagnostic analysis of causes and preventive measures, research has been conducted on the practical use of modern "machine learning" techniques, such as "Deep learning". However, the attention that big companies like Google, Facebook, and Amazon have given to applications for unstructured data and artificial intelligence is not the end of the sector. "This evaluation is poised to catalyze additional exploration into optimal strategies for addressing the multifaceted challenges within the industry by harnessing the computer vision, image processing, and natural language processing capabilities offered by 'Deep learning'. The exploration extends to tackling aspects including the enigma of the black box, ethical considerations and GDPR compliance, cybersecurity intricacies, as well as the cost constraints associated with integrating 'Deep learning'. These aspects are anticipated concerns that construction investigators and practitioners need to consider while embarking on the implementation of such methodologies."

As per the study conducted by, Khan *et al.*, (2020), accurately predicting when the number of people will peak and then start to decline is urgently needed. Professionals in public welfare will benefit much from planning the preventive actions to be done to maintain the nation's economic equilibrium. Python and data visualisation techniques have been used to depict variables like gender, location, age, and others. The outcomes demonstrate that the sigmoid model, a statistical model, outperforms other models. The Sigmoid model also predicts when the curve will begin to flatten and the day on which may anticipate the number of ongoing cases reaching its peak. Data has been transferred into a logarithmic scale using certain feature engineering approaches since it allows for better comparison by removing any extremes or outliers in the data. Our model can be adjusted to forecast lengthy time intervals based on predictions of the short-term interval.

Based on the article which is made by Maulud *et al.*, (2020), for retailers and manufacturers, improved demand forecasting is a competitive advantage, particularly in the fashion, technology, and fresh food industries. This exploratory study highlights how "machine learning" outperforms traditional statistical methods in terms of accuracy when it comes to sales forecasting for

perishable goods with short shelf lives. This leads to better inventory management throughout the chain, lower stock-up rates at the points of sale, better availability for customers, and higher profitability. Business intelligence (BI) is essential in the modern era for formulating strategies and taking informed decisions. Business intelligence is crucial to the unavoidable decision-support mechanism that enables the firm to analyse data at various points throughout business operations. "machine learning" forecasts business demand for the foreseeable future. By comparing the projected data with the actual data and calculating the percentage error, then further test the effectiveness. According to simulation results, accurately estimate demand for the shop with up to 92.38% accuracy by using real-time organisation data and the purposed solution.

Linear regression is conceivably one of the most widespread and complete statistical and machine-learning techniques. Finding a linear relationship between one or more factors is done using linear regression. multiple regression (MLR) and Simple regression are the two variations of linear regression. On the basis of the research conducted by Helmini *et al.*, (2019) The optimum method to maximise prediction and precision is used to compare the performance of polynomial regression and linear regression in this study, which also discusses many works on these topics by various researchers. "A model's effectiveness must be associated with the actual values obtained for the explanatory variables because almost all of the studies reviewed in this review were dataset-focused". Estimating sales in the retail industry before actual sales are known is essential to operating a profitable company. This is because the majority of important judgements must unavoidably be made on these estimates. One of the oldest and most widely used forecasting approaches is the use of statistical forecasting of sales models such as ARIMA (Auto-Regressive Integrated Moving Average). By including features that represent the future that is now known to us, a technique that has not before been explored in state-of-the-art "LSTM"-based forecasting models, further increases the initial "LSTM" model's predictive accuracy.

As per the study of Mosavi *et al.*, (2020), the applications based on deep reinforcement learning (DRL) have become more popular in economics. Reinforcement learning (RL) and "Deep learning" (DL) are just two of the many capabilities that make up DRL, and they both offer a wide range of possibilities for addressing complex dynamic economic systems. DRL can be

characterised by scalability with the possibility to be applied to high-dimensional problems together with noisy and nonlinear patterns of economic data.

In order to give readers of this study a comprehensive understanding of the state-of-the-art, first take a quick look at DL, RL, and deep RL approaches in a variety of economics applications. Moreover, it also can be mentioned that The complexity, resilience, accuracy, performance, computing jobs, risk constraints, and profitability of the DRL architecture as applied to economic applications are highlighted. According to the survey's findings, DRL can deal with actual economic issues in the presence of risk parameters and growing uncertainties with better performance and greater efficiency than standard algorithms.

2.4 Table

Author	Title	Comments
Pallathadka., Ramirez-Asis., Loli-Poma., Kaliyaperumal., Ventayen, and Naved., 2023.	“Applications of artificial intelligence in business management, e-commerce and finance. Materials Today: Proceedings.”	In this article, by utilising a range of fitting algorithms for sales projections, artificial intelligence and data science can increase productivity in lodgings, cafés, dining establishments, online meal delivery services, and restaurants while improving their quality.
Brei., 2020.	“Machine learning in marketing: Overview, learning strategies, applications, and future developments. Foundations and Trends® in Marketing.”	In order to predict the sales of furniture, a public dataset containing a retail store's sales history is inspected. Several forecasting prototypes are used to achieve this goal. First, specific standard time-series forecasting techniques are used, including “Triple Exponential Smoothing and Seasonal Autoregressive Integrated Moving Average (SARIMA)”.

Kumar., Rawat., Mohd, and Husain., 2021.	“Opportunities of artificial intelligence and machine learning in the food industry. Journal of Food Quality, 2021.”	The marketing industry has not yet experienced the extensive effects of artificial intelligence (AI) and machine learning (ML) in many societal sectors. Despite these flaws, ML offers a number of potential advantages, including the chance to use more reliable techniques for the extension of scientific discoveries. This monograph aims at reducing this shortage in four ways.
Javed Awan., Mohd Rahim., Nobanee., Yasin, and Khalaf., 2021.	"A big data approach to black Friday sales. MJ Awan, M. Shafry, H. Nobanee, A. Yasin, OI Khalaf et al.," A big data approach to black Friday sales," Intelligent Automation & Soft Computing."	This study focuses on how artificial intelligence is being used in the food business to reduce human error and increase resource efficiency, as will be discussed in this article. By utilising a range of fitting algorithms for sales projections, artificial intelligence and data science can increase productivity in lodgings, caf��s, dining establishments, online meal delivery services, and restaurants while improving their quality.
Tao., Zhou, and Feeney., 2019.	“ I understand what you are saying: leveraging deep learning techniques for aspect based sentiment analysis.”	This artical focuses on Black Friday sales data from a Kaggle dataset in this study, which has over 550,000 observations and was examined using 10 qualitative and quantitative characteristics”. The name of the category is sales and purchases. Given the continuous nature of the predictor label in this scenario, regression models

		emerge as appropriate choices. Leveraging the MLib "machine learning" library, an integral component of the Apache Spark big data platform, two "machine learning" models were cultivated: linear regression and random forest.
Akinosho., Oyedele., Bilal., Ajayi., Delgado., Akinade, and Ahmed., 2020.	“ Deep learning in the construction industry: A review of present status and future innovations. Journal of Building Engineering,”	The research look in the random forest (RF) networks and “long short-term memory (LSTM)” networks used in "Deep learning" are combined in this paper to present a unique forecasting technique. The suggested approach has an accuracy advantage over previous forecasting techniques since it can handle intricate temporal and regression correlations. On a multivariate dataset from a retailer, tested the novel methodology. And compare the new proposal's predicting performance to that of RF, multiple regression, “ARIMAX”, “LSTM”, and neural networks.
Sengupta., Mugde, and Sharma., 2020.	“Covid-19 pandemic data analysis and forecasting using machine learning algorithms.”	The article uses an Artificial Neural Network (ANN) model was employed to assist vehicle dealers in identifying the many features of cars, including their location, manufacturers, and classification into different categories including "buying," "maintenance," "doors, people, lug_boot,

		safety," and "overall." In order to predict car acceptance, ANN was utilised. The findings demonstrated that the “ANN model had a 99.12% accuracy rate in predicting car approval”.
Tarallo., Akabane., Shimabukuro., Mello, and Amancio., 2019.	“Machine learning in predicting demand for fast-moving consumer goods: An exploratory research. IFAC-PapersOnLine.”	This study helps to assist with the prescriptive and diagnostic analysis of causes and preventive measures, research has been conducted on the practical use of modern "machine learning" techniques, such as "Deep learning". However, the attention that big companies like Google, Facebook, and Amazon have given to applications for unstructured data and artificial intelligence is not the end of the sector.
Khan., Saqib., Alyas., Rehman., Saeed., Zeb., Zareei, and Mohamed., 2020.	“ Effective demand forecasting model using business intelligence empowered with machine learning.”	This research accurately predicting when the number of people will peak and then start to decline is urgently needed. Professionals in public welfare will benefit much from planning the preventive actions to be done to maintain the nation's economic equilibrium. Python and data visualisation techniques have been used to depict variables like gender, location, age, and others.
Maulud, and Abdulazeez., 2020.	“A review on linear regression comprehensive in machine learning. Journal of Applied Science and	This research contributes to the benefit of retailers and manufacturers, offering

	Technology Trends.”	enhanced demand forecasting as a distinct competitive edge, notably within sectors such as fashion, technology, and fresh food. This exploratory investigation underscores the superiority of "machine learning" over conventional statistical techniques in terms of precision, specifically concerning sales forecasting for items with limited shelf lifespans.
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Chapter 3

3. Methodology

CRISP-DM is a widely recognized and robust methodology for guiding the process of data mining and analytics projects. It provides a structured framework comprising a sequence of well-defined steps, ensuring a systematic approach to tackling complex data-related challenges. CRISP-DM encourages iterative development, fostering collaboration between business understanding, data preparation, modeling, evaluation, and deployment stages.

We adopt the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology to guide our Future Sales Prediction project. CRISP-DM's structured approach ensures systematic progress through well-defined stages, from understanding business objectives and data exploration to model deployment and continuous monitoring. By following CRISP-DM, we enhance project organization, facilitate collaboration, and maximize the potential of machine learning and deep learning techniques for accurate and actionable sales forecasts.

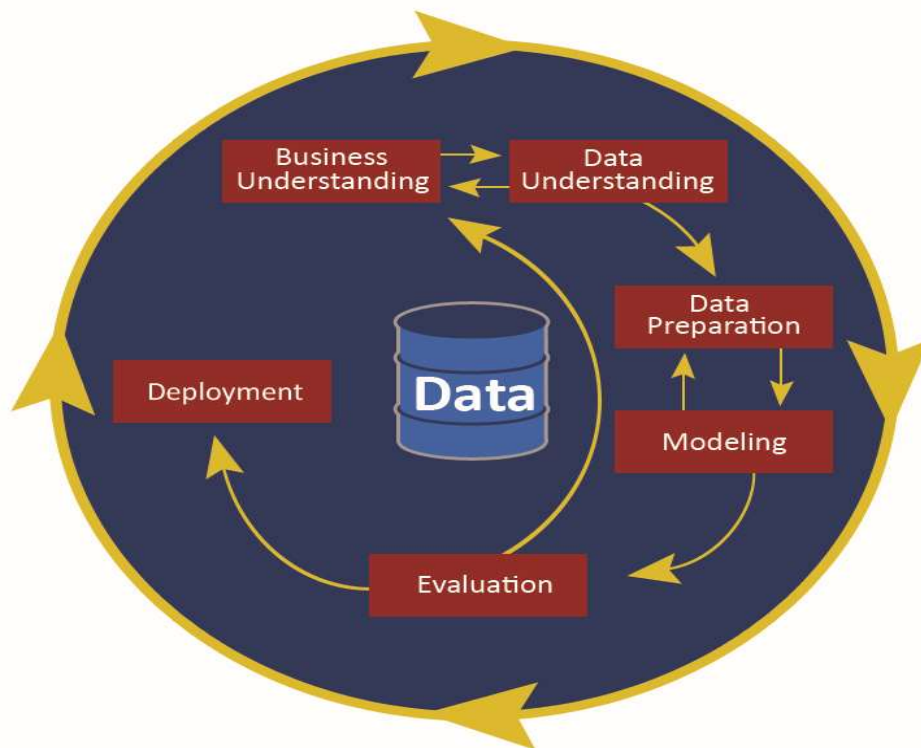


Fig 1: CRISP-DM Methodology

The CRISP-DM steps are as follows:

Business Understanding: This initial phase involves collaborating with stakeholders to establish the project's objectives, requirements, and success criteria. By understanding the business context, one can define clear data mining goals aligned with the organization's needs. Initiate by collaborating with stakeholders to outline project objectives, like precise future sales prediction to optimize resource allocation. Tailor data mining goals based on a thorough grasp of the business context. Ensure project scope aligns with available resources and desired outcomes.

Data Understanding: In this stage, you gather an in-depth understanding of the available data sources. Immerse in sales-related data sources, spanning historical sales records, store attributes, economic indicators, and more. Delve into data distributions, patterns, and anomalies to comprehend dataset intricacies and challenges.

Data Preparation: Cleanse and preprocess data to ensure quality and modeling suitability. Address missing values in sales records and related variables, encode categorical attributes like store type, and engineer features, such as moving averages or lagged sales, capturing temporal dependencies.

Modeling: Opt for suitable machine learning and deep learning techniques to forecast future sales. Experiment with algorithms like regression, time series models, and neural networks (e.g., LSTM). Train models using prepared data, fine-tuning hyperparameters for optimal performance.

Evaluation and Deployment : Assess model accuracy with various metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE). Roll out the selected model to the business environment. Document user guidelines for leveraging the model in sales prediction. Devise a seamless deployment strategy for smooth integration, enabling stakeholders to harness accurate sales forecasts.

In this study, data analysis is conducted to uncover future sales patterns. The following are the essential prerequisites needed for this analysis:

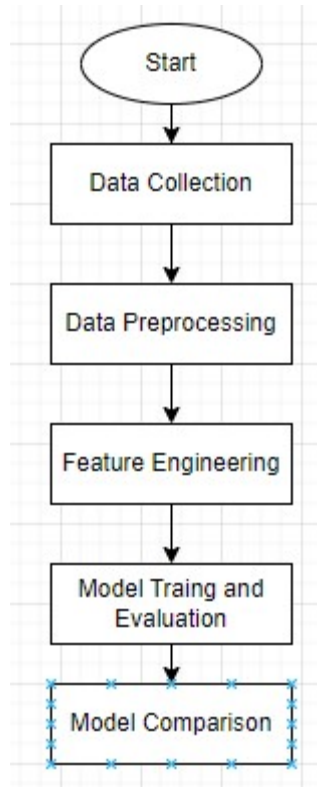


Fig 2: Project Methodology

Project Requirement: To meet the demands of the project, Google Colab was employed as the software platform for code development and execution. Offering a user-friendly interface akin to Anaconda Navigator, Google Colab eliminates the need for installation and facilitates seamless file sharing.

- a) **Python:** Python is a versatile and widely-used programming language known for its simplicity and readability. It serves as the foundation for various machine learning and deep learning libraries, making it a preferred choice for data analysis and model development.
- b) **scikit-learn (sklearn):** scikit-learn is a robust machine learning library in Python. It provides a wide range of tools for tasks such as classification, regression, clustering, and dimensionality reduction. Its user-friendly interface and extensive documentation make it a go-to library for implementing various machine learning algorithms.

- c) **Matplotlib:** Matplotlib is a powerful data visualization library for Python. It offers a variety of plotting functions to create graphs, charts, and figures, making it essential for presenting and understanding data patterns, trends, and relationships visually.
- d) **Keras:** Keras is a high-level neural network library built on top of lower-level frameworks like TensorFlow. It simplifies the process of designing and training neural networks by providing a user-friendly API. Keras is well-suited for rapid prototyping and experimentation with deep learning architectures.
- e) **Model Used:** Attention Mechanisms, Graph Neural Networks, Recurrent Neural Networks, Long Short-Term Memory, Autoregressive Integrated Moving Average and Vector Autoregression

3.1 Dataset

The dataset chosen for the thesis report has a size of around 49.3 megabytes (MB) and comprises a total of 374,247 records, distributed across 21 distinct columns. The dataset was acquired from the provided source, <https://data.world/revanthkrishnaa/amazon-uk-sales-forecasting-2018-2021>.

The columns within the dataset encompass a wide range of attributes essential for robust sales forecasting. These include key variables such as 'Date', 'Store', 'Dept', 'Weekly_Sales', 'Type', 'Size', 'Temperature', 'Fuel_Price', 'CPI', 'Unemployment', 'IsHoliday', 'Year', 'Month', 'Week', 'max', 'min', 'mean', 'median', 'std', and 'Total_MarkDown'. This assortment of attributes covers temporal dimensions, store and department identification, economic metrics, weather-related factors, and information about markdowns.

3.2 Data Analysis using tableau

Dashboard 1:

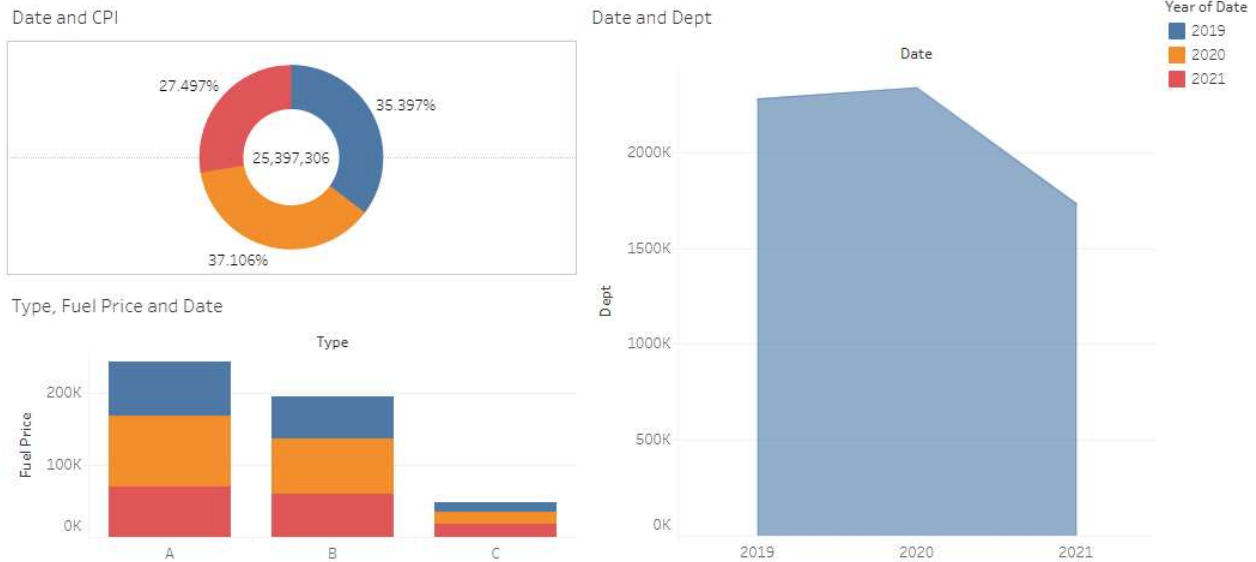


Fig 3: Dashboard 1

In the above dashboard 1st worksheet shows the donut for date and CPI, here 2020 has highest CPI, The second worksheet show the area chart for date and dept , here 2020 has the highest number of dept. The third worksheet show the stacked bar for type, fuel price and date, here A type has the highest count of fuel price as compare to other type B and C.

Dashboard 2:



Fig 4: Dashboard 2

In the above dashboard, the first worksheet shows the treemap for date, history, and fuel price. Here, 2019 has the maximum special holiday week as compared to 2020 and 2021. The second worksheet shows the bar chart for date, max, and min; here, 2020 has a maximum count of

MAX. The third worksheet shows the stacked bar chart for date, mean, and type; here, 2020 has the maximum count of the mean.

Dashboard 3:

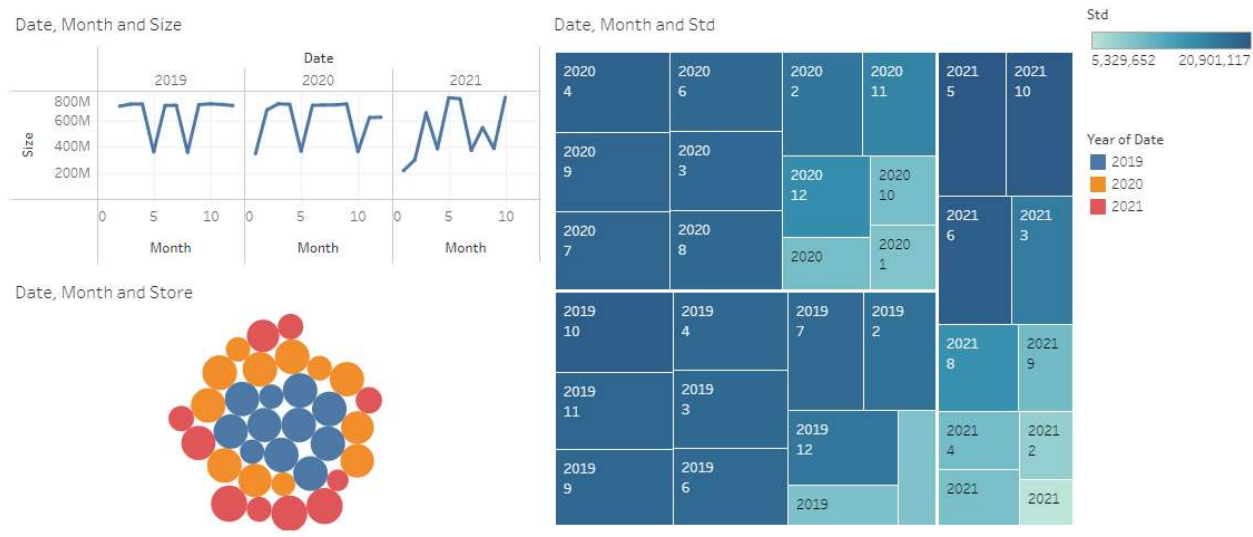


Fig 5: Dashboard 3

In the above dashboard, the first worksheet shows the line chart for date, month, and size; here, month is on the x axis and size is on the y axis. The second worksheet shows the treemap for date, month, and standard deviation; here, the dark region has a higher standard deviation than the light region. The third worksheet shows the bubble chart for date, month, and store; here different colours show the different dates, like 2020, 2019 and 2021.

Dashboard 4:

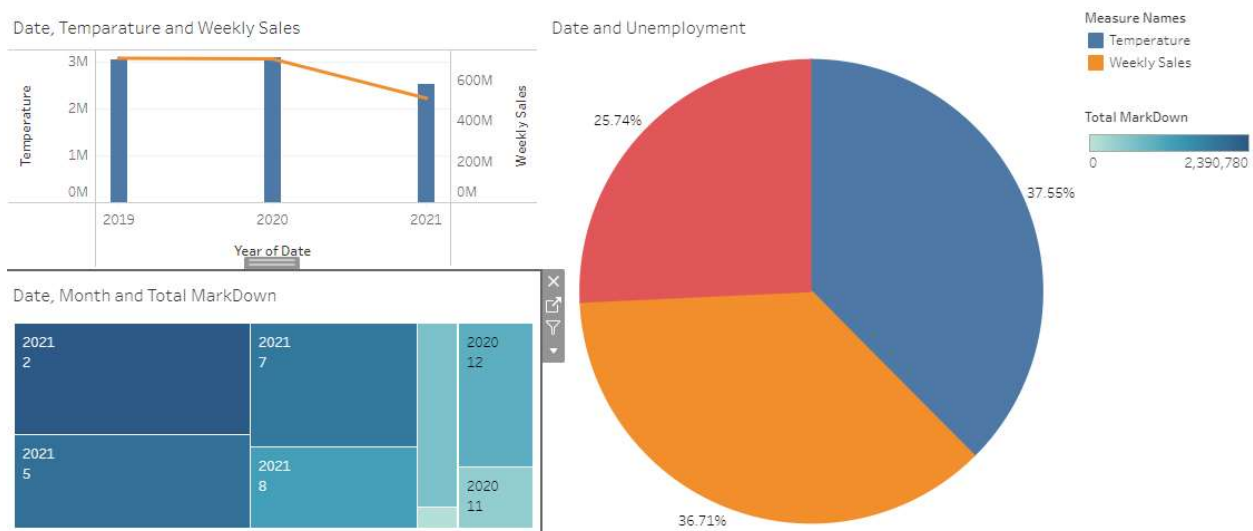


Fig 6: Dashboard 4

In the above dashboard, the first worksheet shows the line chart for date, month, and size; here, month is on the x axis and size is on the y axis. The second worksheet shows the treemap for date, month, and standard deviation; here, the dark region has a higher standard deviation than the light region. The third worksheet shows the bubble chart for date, month, and store; here different colours show the different dates, like 2020, 2019 and 2021.

Dashboard 5:

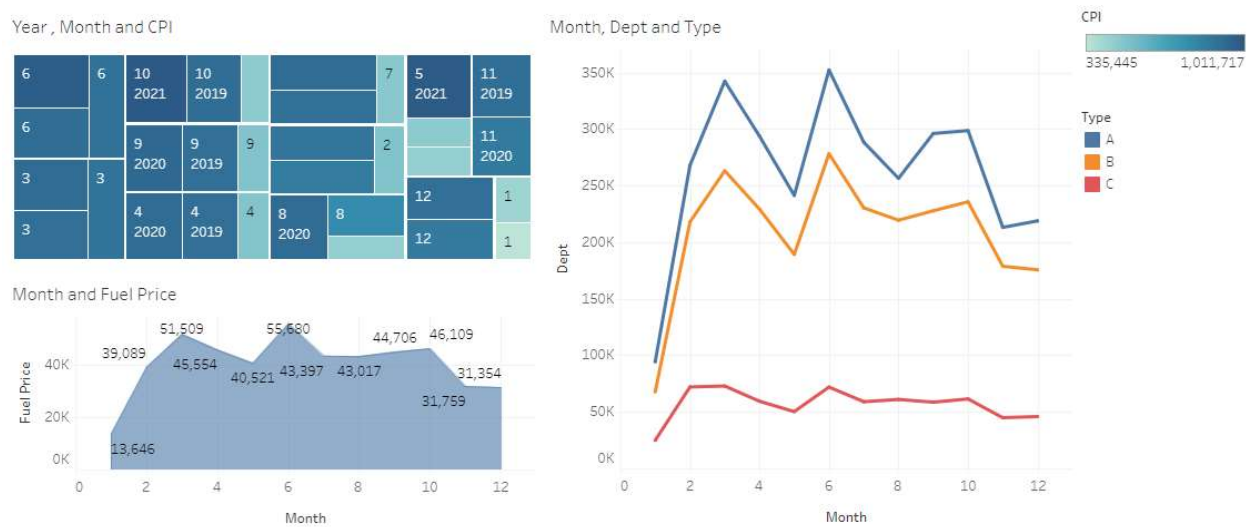


Fig 7: Dashboard 5

In the above dashboard, the first worksheet shows the treemap for year, month, and cpi. Here, the dark region has the highest count of cpi as compared to the light region. The second worksheet shows the line chart for month, department, and type; different colours indicate the different types. The third worksheet shows the area plot for month and fuel price; here, month 6 has the maximum count of fuel price.

Dashboard 6:

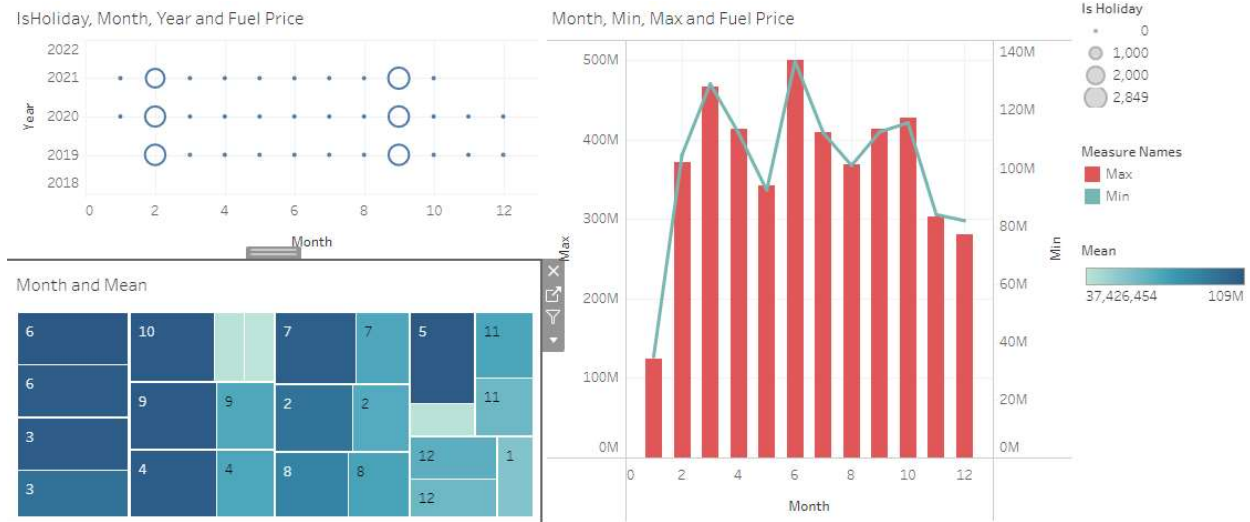


Fig 8: Dashboard 6

In the above dashboard, the first worksheet shows a scatterplot for IsHoliday, Month, Year, and Fuel Price. Here, the small circle has the lowest holiday as compared to the large circle. The second worksheet shows the visualisation for month, min, max, and fuel price; here, red means max and blue means mean. The third worksheet shows the treemap for month and mean; here, the dark region has a higher count of mean as compared to the light region.

Dashboard 7:

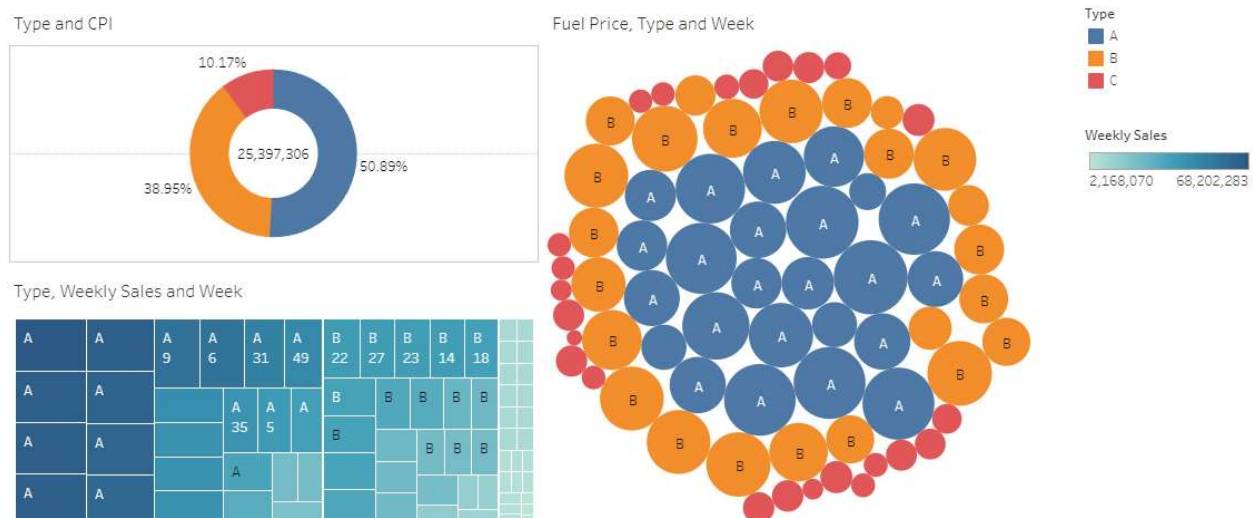


Fig 9: Dashboard 7

In the above dashboard, the first worksheet shows the donut for type and CPI; here, type A has the maximum percentage of CPI as compared to types B and C. The second worksheet shows the bubble chart for fuel price, type, and week; different colours indicate the different types. The

third worksheet shows the treemap for type, weekly sales, and week. Here, the dark region has the maximum number of weekly sales as compared to the light region.

CONCLUSION:

This report features visualizations created using Tableau, employing a range of chart types such as bar charts, pie charts, and bubble charts. The Amazon sales dataset serves as the foundation for these visualizations, utilizing its various attributes for insightful representation. By harnessing the dataset's diverse attributes, this study produces meaningful visualizations that effectively communicate information and trends.

3.3 Data Pre-processing

Prior to conducting data pre-processing, it is essential to import the necessary libraries and the dataset using the pandas library.

```
#import all needed libraries
import numpy as np
import pandas as pd
import itertools
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import IsolationForest
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, mean_absolute_percentage_error
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.vector_ar.var_model import VAR
import math
import keras
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import LSTM
from keras.layers import Dropout
from keras.layers import *
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_absolute_error
from keras.callbacks import EarlyStopping
import tensorflow as tf
from tensorflow.keras.models import Sequential
#from tensorflow.keras.layers import Dense, LSTM
import matplotlib.pyplot as plt
import statsmodels.api as sm

import warnings
warnings.filterwarnings('ignore')
```

Fig 10: Importing the libraries

In preparation for data preprocessing, a comprehensive suite of libraries has been imported. Numpy facilitates intricate mathematical computations, while Pandas empowers seamless data manipulation. To visualize and glean insights, Matplotlib and Seaborn come into play, synergizing to deliver compelling graphical representations. Building on these visualizations, graph analysis is enabled by leveraging the imports from the previous section. Recognizing the significance of maintaining a streamlined workflow, the warnings library has been introduced to effectively manage and minimize warning occurrences. Notably, the incorporation of sklearn equips us with advanced machine learning capabilities, while Keras integration enables the development and training of intricate deep learning models. This cohesive integration of libraries collectively amplifies our analytical prowess, paving the way for comprehensive data preprocessing and insightful analysis.

```
#read the dataset
df = pd.read_csv("amazon_sales_dataset.csv")
df.head()
```

Fig 11: Read the dataset

Subsequently, the dataset is imported using the read method from the pandas library. Given that our dataset is in a CSV format, we employ the read_csv function. It's important to note that datasets could also be stored as Excel or JSON files. To verify successful importation, the df.head function is utilized to present the initial five records from the dataset. This action serves to confirm the dataset's incorporation. The following figure will illustrate this initial dataset snapshot.

```
#shape of dataset
df.shape|
(374247, 21)
```

Fig 12: shape of dataset

The shape of the DataFrame, represented as (374247, 21), indicates that it contains 374,247 rows and 21 columns.

```
#Statistical description
df.describe()
```

Fig 13: statistical description of dataset

The **df.describe()** function provides a concise statistical overview of the DataFrame's numerical columns. It presents essential statistics including count, mean, standard deviation, minimum, 25th percentile, median (50th percentile), 75th percentile, and maximum values for each column. This summary aids in understanding the central tendency, spread, and distribution of the data, enabling initial insights into the dataset's characteristics.

```
#convert datae to date time
df['Date'] = pd.to_datetime(df['Date'])
df_monthly = df.groupby(pd.Grouper(key='Date', freq='M')).agg({'Weekly_Sales': 'sum'}).reset_index()
```

Fig 14: Convert Date to date time

The 'Date' column is transformed into a datetime format using the **pd.to_datetime()** function. Subsequently, the dataset is aggregated on a monthly basis using the **groupby()** function with a frequency of 'M' (indicating months). The 'Weekly_Sales' values are summed within each month, resulting in a new DataFrame **df_monthly** that captures the total sales for each month.

3.4 Data Visualisation and Feature Engineering

Data visualization is a vital component of data analysis that transforms complex datasets into clear, intuitive visual representations. Through charts, graphs, and other visual elements, data visualization simplifies the communication of trends, patterns, and insights hidden within raw data. It aids in uncovering correlations, outliers, and anomalies, making it easier to comprehend and interpret information. Effective data visualization enhances decision-making by presenting complex concepts in a digestible format, enabling stakeholders to gain valuable insights, identify opportunities, and formulate strategies. By bridging the gap between data and understanding, data visualization plays a pivotal role in extracting meaningful knowledge from datasets and driving informed actions.

As part of the data preprocessing process, several key steps have been undertaken to enhance the dataset's suitability for analysis and modeling. Notably, temporal features have been extracted, encompassing attributes like month, year, and potential seasonality patterns. This extraction facilitates the incorporation of time-related insights into the analysis, enabling the identification of trends and patterns that evolve over different time intervals.

Furthermore, lagged features have been calculated, capturing historical information by including past sales and item prices from preceding months. These lagged attributes introduce a temporal context to the data, enabling the analysis to account for the influence of previous periods on the

current sales landscape. This strategic inclusion of lagged features contributes to capturing the inherent time-dependent dependencies and correlations within the dataset, thereby enhancing the potential accuracy and efficacy of subsequent modeling and forecasting endeavors.

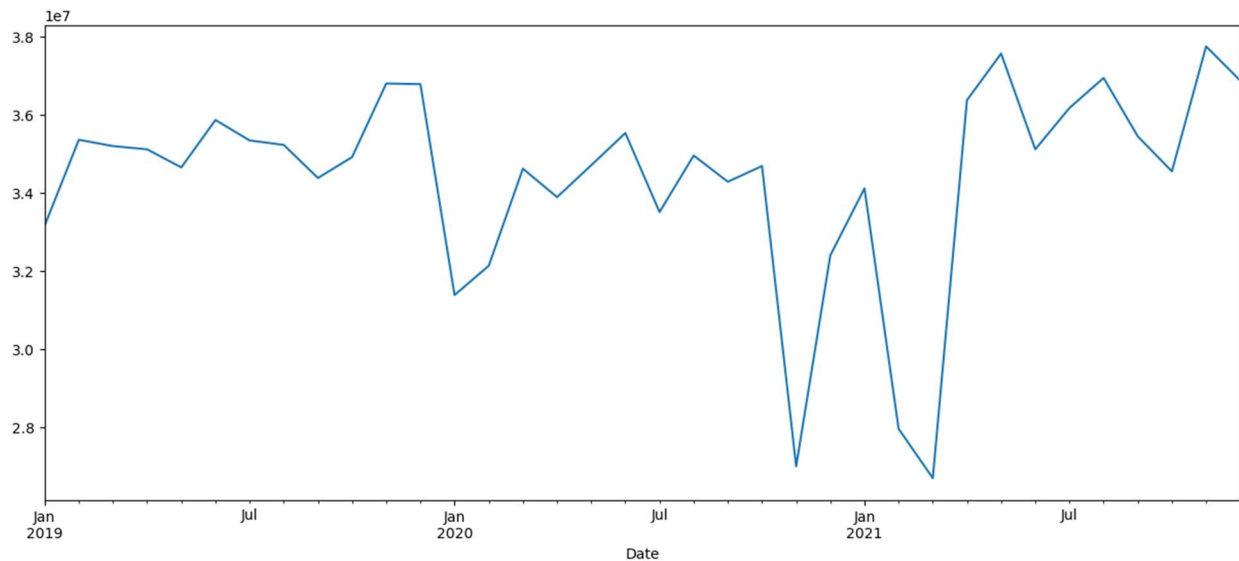


Fig 15: Plot for y

The code snippet creates a plot of the average weekly sales (**Weekly_Sales**) using data resampled on a monthly frequency ('MS'). The focus is on the year 2021. The resulting plot visualizes the sales trends over this specific year, offering a concise overview of sales patterns.

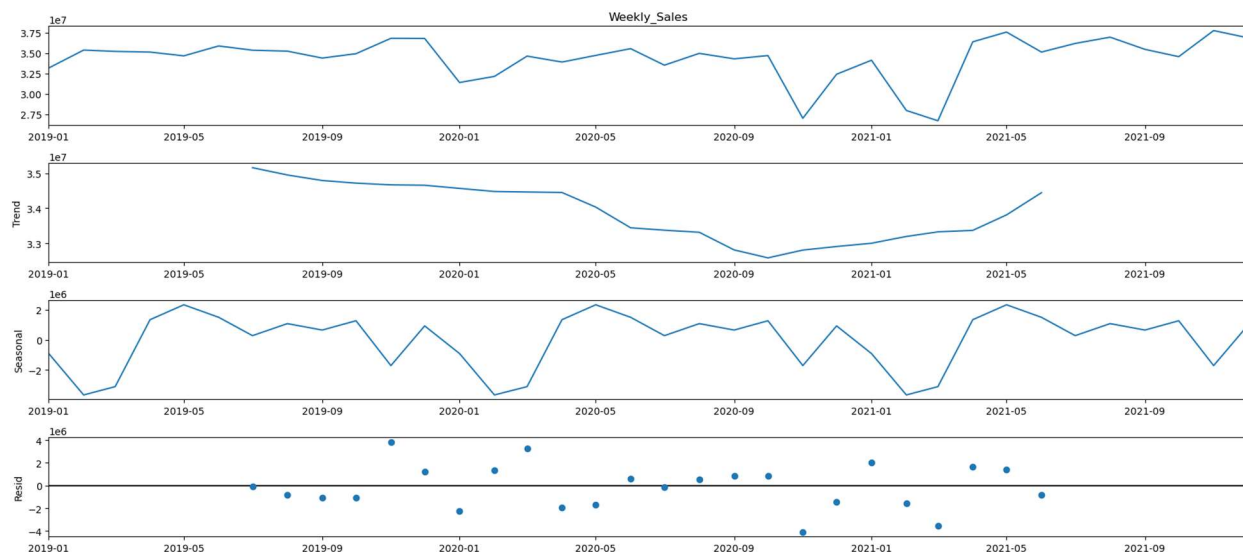


Fig 16: Seasonal, resid and trend

This decomposition aids in identifying patterns, irregularities, and fluctuations within the dataset, contributing to a more comprehensive understanding of the time series dynamics.

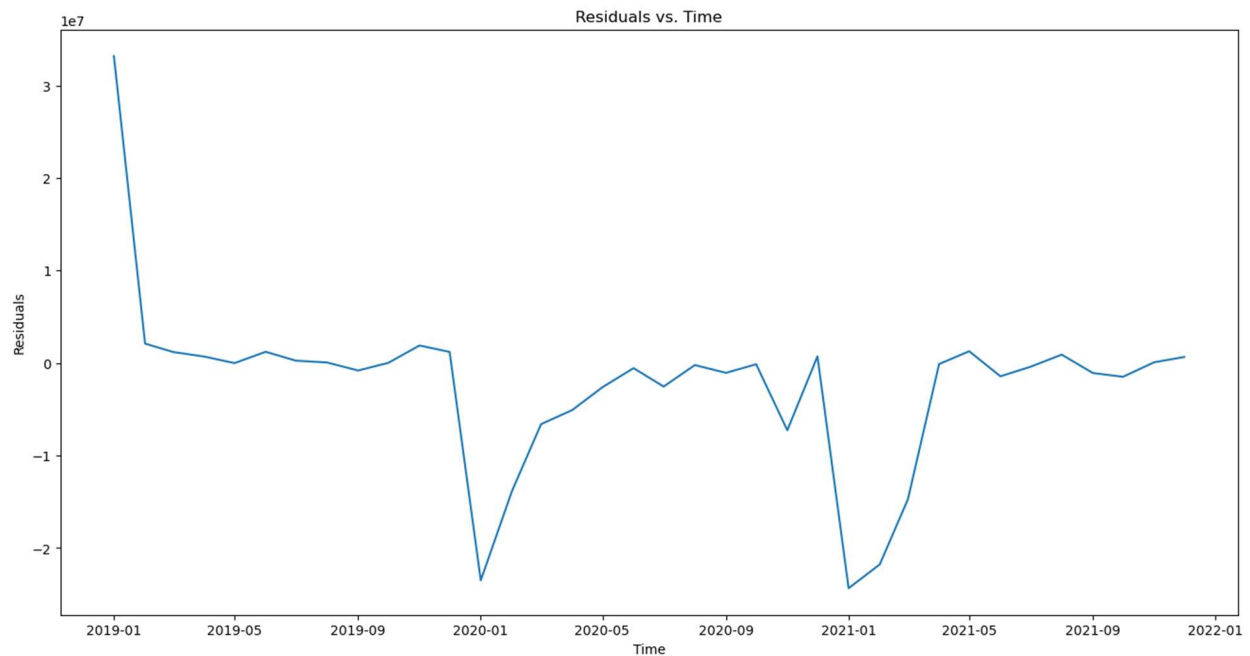


Fig 17: Residuals vs Time

This visual representation serves to examine the dispersion of residuals across time intervals. By plotting the residuals over time, any discernible patterns or trends that might exist become apparent, shedding light on the temporal behavior of the residuals within the dataset.

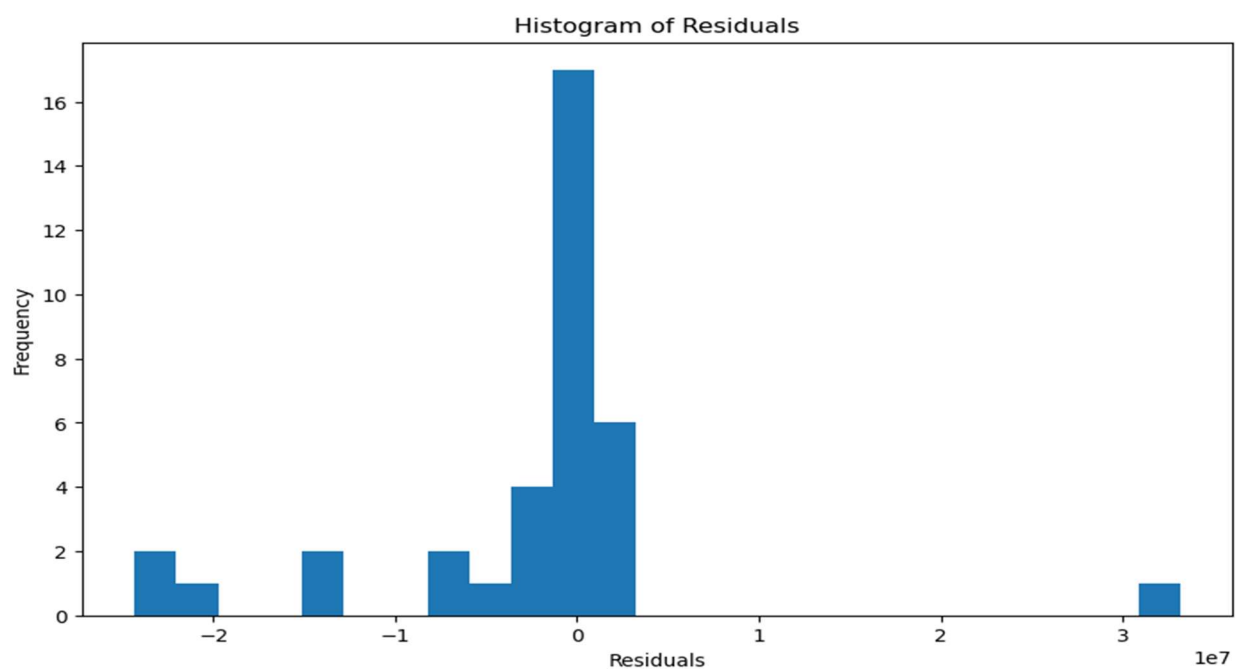


Fig 18: Histogram of Residuals

The histogram aids in comprehending the frequency distribution of the residuals. By depicting the occurrence of different residual values, this graphical tool offers insights into their overall distribution characteristics, which can indicate the presence of skewness, symmetry, or outliers.

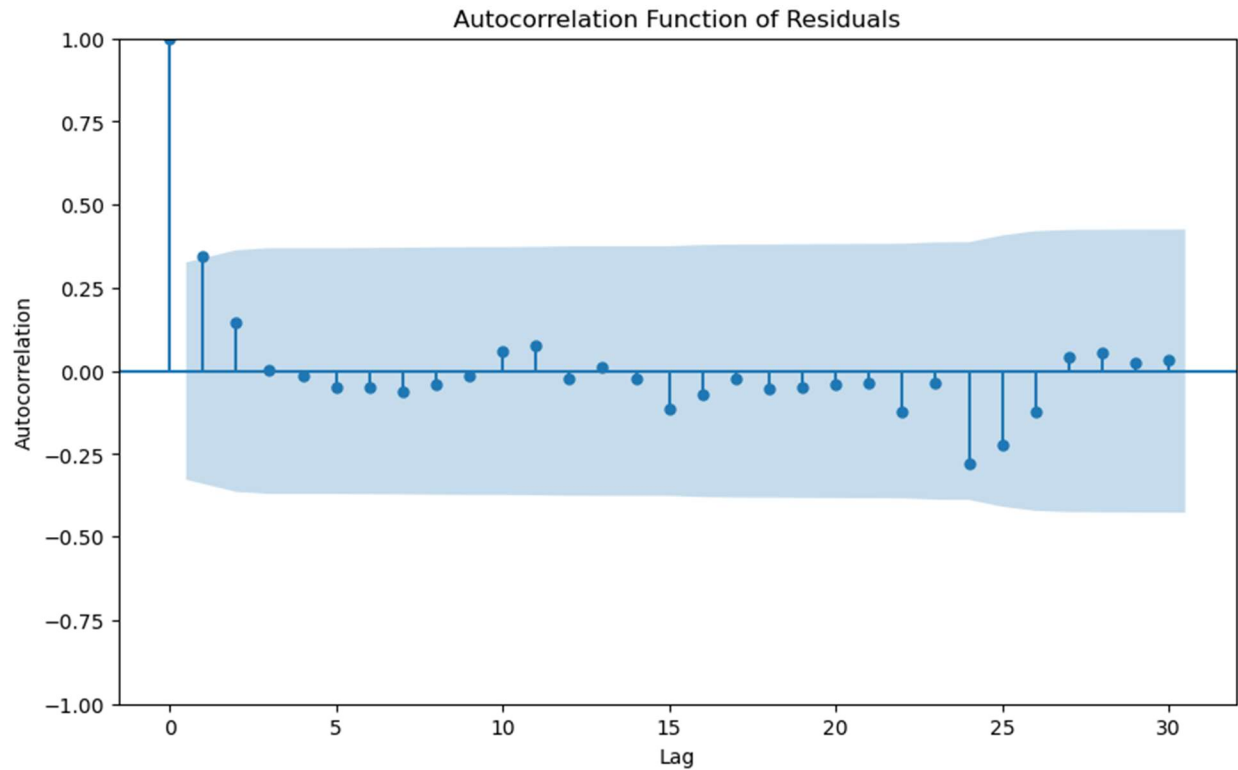


Fig 19: Autocorrelation plot

This plot gauges the correlation between residuals at varying time lags. By evaluating how residuals at different points in time correlate with each other, the ACF plot assists in identifying any potential autocorrelation patterns that might persist within the residuals.

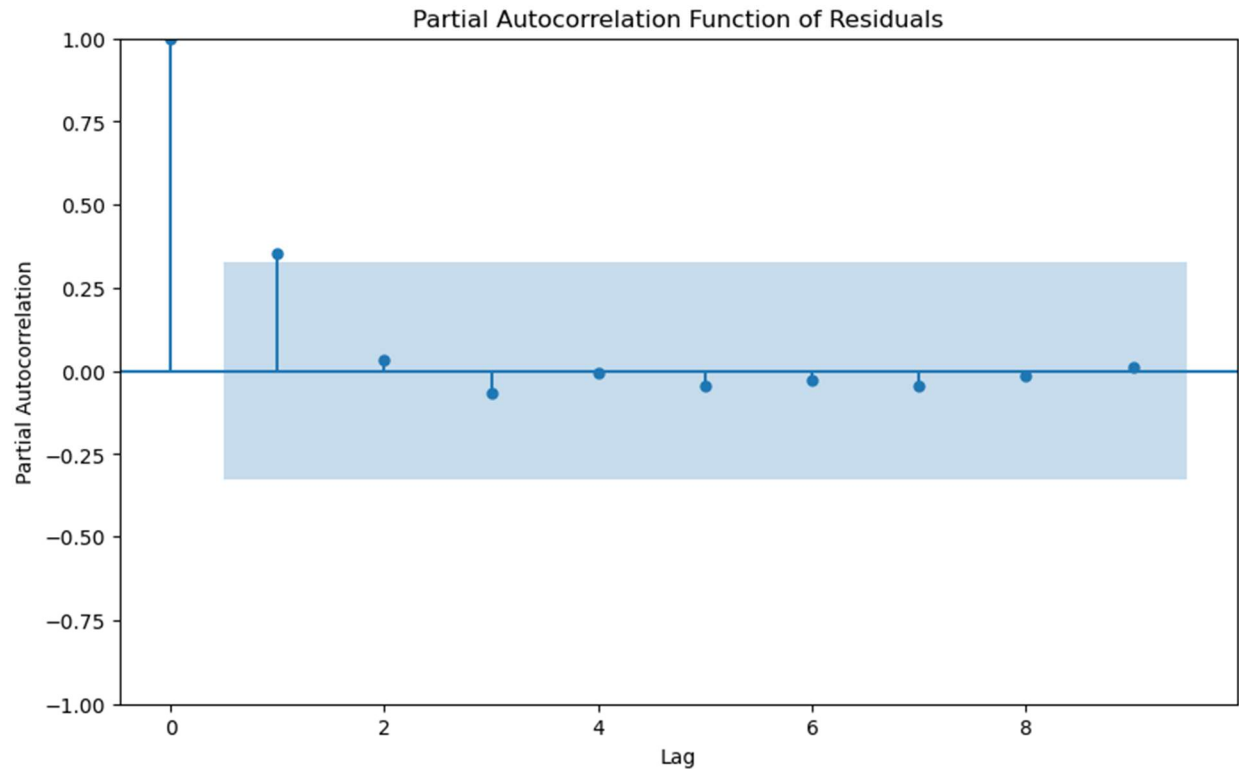


Fig 20: Partial Autocorrelation Function

The PACF plot highlights the correlation between residuals at various lags while considering the influence of intermediate lags.

```
# Scale the data
scaler = MinMaxScaler(feature_range=(0,1))
scaled_data = scaler.fit_transform(dataset)
```

Fig 21: Standard scaler

To ensure uniformity in the data and facilitate accurate modeling, the data is scaled using the **MinMaxScaler**. This scaling process transforms the data into a specific range, typically between 0 and 1, preventing larger values from disproportionately affecting the analysis. The **scaler** object is created, employing the **MinMaxScaler** class with the desired **feature_range** of (0,1). The **fit_transform** method is then employed to apply the scaling transformation to the **dataset**, yielding **scaled_data**. This operation is fundamental for enhancing the compatibility and effectiveness of subsequent analytical and modeling procedures.

3.6 Model Training

The pivotal phase of model training encompasses a series of essential steps to equip the chosen algorithm with the capability to make accurate predictions. Commencing with the preparation of training data, which includes organizing input features and target values, the process swiftly advances to the initialization of the selected machine learning or deep learning model, complete with its distinctive hyperparameters. Subsequently, the model embarks on an iterative learning journey, steadily refining its internal parameters to minimize the discrepancy between projected and actual target values. This process hinges on optimization techniques like gradient descent, and is governed by defined epochs and batch sizes, regulating the number of iterations and the data quantity processed in each step. Crucially, a validation dataset comes into play, evaluating the model's performance beyond the training data, thus gauging its capacity to generalize accurately. As training progresses, the trajectory of metrics such as loss and accuracy across epochs furnishes a comprehensive overview of the model's convergence and performance trends.

```
# Scale the dataset
scaler = MinMaxScaler(feature_range=(0, 1))
dataset = scaler.fit_transform(dataset)

# Split the dataset into train and test
train_size = int(len(dataset) * 0.80)
train, test = dataset[0:train_size, :], dataset[train_size:len(dataset), :]

# Create input-output pairs
def create_dataset(dataset, look_back=1):
    X, Y = [], []
    for i in range(len(dataset) - look_back):
        a = dataset[i:(i + look_back), 0]
        X.append(a)
        Y.append(dataset[i + look_back, 0])
    return np.array(X), np.array(Y)

# Define the Look_back parameter
look_back = 28

# Create the training dataset
X_train, Y_train = create_dataset(train, look_back)

# Create the test dataset
X_test, Y_test = create_dataset(test, look_back)
```

Fig 22: split the dataset into training and testing

The dataset is divided into training and test subsets, with approximately 80% allocated for training and the remainder for testing. To enable sequence-based forecasting, input-output pairs are constructed through a custom function. This function, named **create_dataset()**, extracts segments of historical data to create inputs and corresponding outputs, effectively framing the problem as a sequence prediction task. This approach facilitates the utilization of previous time steps to predict future values, a fundamental concept in time series forecasting.

```
# Build the LSTM model
model = Sequential()
model.add(LSTM(128, return_sequences=True, input_shape= (x_train.shape[1], 1)))
model.add(LSTM(64, return_sequences=False))
model.add(Dense(25))
model.add(Dense(1))

# Compile the model
model.compile(optimizer='adam', loss='mean_squared_error')

# Train the model
model.fit(x_train, y_train, batch_size=1, epochs=1)
```

Fig 23: Build the LSTM

Long Short-Term Memory (LSTM) Networks : Long Short-Term Memory (LSTM) Networks are a pivotal advancement within the realm of recurrent neural networks, specifically engineered to surmount the limitations of traditional RNNs in capturing long-range dependencies. LSTMs have emerged as a potent solution for sequence modelling and prediction, finding applications in diverse fields such as natural language processing, speech recognition, and time series analysis.

LSTMs tackle the vanishing gradient issue by incorporating memory cells and gating mechanisms. These components regulate the flow of information, enabling the network to retain important context over extended sequences. This property proves vital for preserving temporal patterns, handling gaps in data, and modelling relationships between distant time steps.

In language modeling, LSTMs exhibit a remarkable capacity to understand and generate coherent text by considering the context of preceding words. Similarly, in speech recognition, they decipher spoken language patterns, making them valuable for applications like virtual assistants.

However, LSTMs are not without challenges. Their increased complexity can lead to overfitting, and they require substantial computational resources for training. Furthermore, while LSTMs excel in capturing short-term dependencies, very long sequences can still pose challenges.

In essence, LSTMs have revolutionized sequence-based tasks by enabling models to understand and learn from extensive sequences of data. Their ability to capture intricate temporal dependencies positions them as a pivotal tool in deciphering sequential patterns and predicting future values.

Attention Mechanisms : Attention Mechanisms have heralded a transformative era in machine learning, imparting an unprecedented level of precision to diverse applications. Especially prominent in natural language processing and image analysis, attention mechanisms empower models to concentrate selectively on specific input elements, improving their predictive capabilities. This is achieved by assigning varying degrees of importance to different parts of the input data, effectively enabling models to focus on contextually relevant information while downplaying irrelevant details.

In the realm of machine translation, attention mechanisms revolutionize the understanding and translation of source sentences into target languages. By discerning and emphasizing pertinent segments of the source sentence, the model crafts more accurate and coherent translations, accounting for the nuances of context and language structure. Attention mechanisms excel in capturing long-range dependencies and intricate relationships, effectively enhancing the model's ability to generate precise outputs.

Beyond natural language processing, attention mechanisms have found applicability in image analysis. These mechanisms enable models to identify and emphasize salient regions within images, leading to advancements in tasks like object detection, image captioning, and visual question answering. This fine-grained focus on relevant areas of the input has significantly elevated the accuracy and interpretability of image-based models.

In essence, attention mechanisms's ability to adapt to varying context and assign significance to specific features empowers machine learning models to make more informed predictions. This paradigm shift has amplified the effectiveness of various applications, underscoring the significance of attention mechanisms in modern artificial intelligence.

Graph Neural Networks (GNNs) : Graph Neural Networks (GNNs) represent an evolution in machine learning, tailored to address data organized in graph structures. With applications across diverse domains such as social networks, recommendation systems, and biology, GNNs excel in capturing intricate relationships within complex networks.

GNNs operate through iterative information propagation, allowing nodes to accumulate insights from their neighbors. This dynamic process enables GNNs to comprehend both the individual attributes of nodes and the broader network topology. As a result, they excel in tasks like node classification, where nodes are assigned labels based on their features and connections.

Moreover, GNNs are adept at link prediction, forecasting future connections within a graph.

In social network analysis, GNNs reveal influential nodes and communities, aiding in understanding information dissemination dynamics. Within recommendation systems, GNNs provide personalized recommendations by leveraging users' social connections and preferences.

In the field of biology, GNNs decode molecular structures, enabling breakthroughs in drug discovery and protein interaction analysis.

Despite their effectiveness, GNNs encounter challenges related to scalability, over-smoothing, and generalization to dynamic graphs. Researchers are actively addressing these issues to further refine GNNs' capabilities. Nevertheless, GNNs' aptitude for modeling complex relationships and dependencies in graph-structured data ensures their enduring relevance in modern machine learning. Their role in unveiling latent patterns and interconnections marks them as an invaluable asset for comprehending intricate relationships in a diverse array of datasets.

Recurrent Neural Networks (RNNs) : Recurrent Neural Networks (RNNs) have emerged as a cornerstone of sequence modeling, offering unparalleled capabilities in processing time-dependent data. Their design allows them to retain memory of past inputs, making them particularly suitable for tasks like natural language processing, speech recognition, and time series forecasting.

RNNs work by maintaining an internal state that evolves as new inputs are processed. This internal memory facilitates the capture of temporal dependencies and patterns, enabling the network to remember information from earlier time steps. However, conventional RNNs have limitations in capturing long-range dependencies due to vanishing gradient problems, which can impede the flow of gradients during training.

To address this, more advanced RNN variants like Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) have been introduced. LSTMs, in particular, incorporate memory cells and gating mechanisms that enable them to capture and retain information over extended sequences. This makes them highly effective in tasks such as language modeling, where understanding the context of previous words is crucial for generating coherent text.

Despite their efficacy, RNNs face challenges in modeling very long sequences and dealing with noisy or incomplete data. Additionally, they can be computationally expensive due to their sequential nature. Nevertheless, RNNs remain indispensable for understanding and modeling sequences, and they serve as foundational components in many advanced architectures.

Autoregressive Integrated Moving Average (ARIMA) : Autoregressive Integrated Moving Average (ARIMA) models are a foundational framework for time series analysis and forecasting. Combining autoregressive (AR) and moving average (MA) components with differencing (I), ARIMA models are adept at capturing various temporal patterns and seasonality inherent in time-dependent data.

ARIMA models are particularly effective in scenarios where historical observations influence future values. The autoregressive component considers previous values of the series, while the moving average component captures the impact of past forecast errors. The differencing step addresses trends and seasonality by transforming the data into a stationary form.

These models are widely utilized in economic forecasting, financial analysis, and demand prediction. In economic forecasting, ARIMA models help anticipate future trends in indicators like GDP and inflation. In finance, they aid in predicting stock prices and volatility. In retail, ARIMA models are employed to forecast sales and optimize inventory management.

However, ARIMA models have certain limitations. They assume that relationships between variables are linear and stationary, which might not hold true for all datasets. Additionally, ARIMA models may struggle with complex patterns that deviate from simple autoregressive or moving average behavior.

In summary, ARIMA models provide a versatile and interpretable framework for analyzing time series data. Their capability to capture temporal patterns and forecast future values positions them as a reliable tool for various forecasting tasks, while their intuitive structure makes them accessible to a broad audience.

Vector Autoregression (VAR) : Vector Autoregression (VAR) models represent a critical advancement in multivariate time series analysis, allowing for the exploration of relationships and interactions between multiple variables. Building on the principles of autoregression, VAR models extend the concept to encompass multiple variables, making them suitable for understanding dynamic systems with interdependencies.

In VAR models, each variable is expressed as a linear combination of its past values and the past values of other variables. This enables the modeling of complex relationships and feedback loops between variables, uncovering intricate patterns that might not be apparent through univariate analysis.

VAR models find extensive use in economics, finance, and macroeconomic forecasting. In economics, they help analyze the dynamic interplay between economic indicators, revealing insights into factors affecting variables like employment, inflation, and consumer spending. In finance, VAR models assist in modeling correlations and contagion between financial assets.

Despite their strengths, VAR models assume linear relationships and stationary data, which might not hold in all scenarios. They are also sensitive to model specification and may require careful consideration of lag orders.

In essence, VAR models provide a powerful framework for understanding interdependencies among multiple time series variables. By accounting for mutual influence and capturing dynamic relationships, VAR models offer a nuanced perspective on complex systems and contribute to informed decision-making in various domains.

Chapter 4

4. RESULT AND MODEL EVALUATION

For the Status target Variable:

Upon applying multiple machine learning models to predict the target variable "Weekly Sales," the outcomes are summarized as follows:

Model	MSE	MAPE
ARIMA	1.076e+14	18.76%
LSTM	1.902e+13	8.35%
Attention Mechanism	9.503e+13	24.90%
GNN	0.000480	2.11%
RNN	2.093e+13	12.39%
VAR	8.662e+12	7.83%

Table 2: Result of Modes

- **Mean Squared Error (MSE):** MSE measures the average squared difference between the predicted and actual values. By squaring the errors, it accentuates larger discrepancies and thereby penalizes outliers more heavily. This sensitivity to outliers could make MSE less robust to extreme values. A lower MSE signifies a better fit of the model to the data, as it suggests that the model's predictions are, on average, closer to the actual values.
- **Root Mean Squared Error (RMSE):** RMSE is the square root of the MSE, and it provides a clearer sense of the average magnitude of the prediction errors in the original units of the target variable. This makes it more interpretable and relatable to the actual values being predicted. Like MSE, a lower RMSE indicates superior model performance in terms of prediction accuracy.
- **Mean Absolute Percentage Error (MAPE):** MAPE calculates the percentage difference between the predicted and actual values, and then takes the average of these percentage errors. It's a measure of relative prediction accuracy, which is particularly useful when you want to understand the model's performance in relation to the scale of the actual values. MAPE is expressed in percentage terms, making it intuitive to understand. A lower MAPE suggests that the model's predictions, on average, have smaller proportional deviations from the true values.

Result Analysis for Each Model:

ARIMA:

The ARIMA model demonstrates poor performance, evidenced by a high MSE of $1.076e+14$ and a corresponding RMSE of 10,374,171.31. The MAPE of 18.76% indicates that, on average, the model's predictions deviate by almost 19% from the actual values. This suggests that ARIMA struggles to capture the underlying patterns and dynamics of the data, leading to less accurate predictions.

LSTM:

The LSTM model outperforms ARIMA with significantly lower MSE ($1.902e+13$) and RMSE (4,361,673.24) values. The lower MAPE of 8.35% suggests that the LSTM's predictions are, on average, closer to the actual values compared to ARIMA. LSTM's ability to capture temporal dependencies and trends contributes to its better predictive capabilities.

Attention Mechanism:

The Attention Mechanism model presents challenges, as indicated by its high MSE of $9.503e+13$ and a relatively high MAPE of 24.90%. Unfortunately, the RMSE value is not provided. The substantial MAPE implies notable percentage errors in predictions, signifying that the attention mechanism might not effectively capture relevant information within the data.

GNN:

The GNN model excels in predictive performance with an exceptionally low MSE (0.000480) and RMSE (0.0219). The MAPE of 2.11% suggests that GNN's predictions have only a small average percentage error. GNN's capability to capture complex relationships within data, especially in graph-like structures, contributes to its remarkable performance.

RNN:

The RNN model's performance is similar to LSTM's, with moderately high MSE ($2.093e+13$) and RMSE (4,574,596.38) values. The MAPE of 12.39% indicates that the RNN's predictions have a moderate average percentage error. While RNN captures some temporal patterns, there's room for improvement.

VAR:

The VAR model showcases competence with a low MSE (8.662×10^{12}) and RMSE (2,943,177.61). The MAPE of 7.83% suggests that VAR's predictions are, on average, close to the actual values. VAR's effectiveness in capturing relationships between variables contributes to its favorable performance.

In summary, the GNN and VAR models demonstrate strong predictive capabilities, while LSTM and RNN also perform reasonably well. ARIMA and the Attention Mechanism model lag behind due to their challenges in capturing data patterns effectively. The choice of the most suitable model depends on the specific data characteristics and prediction goals.

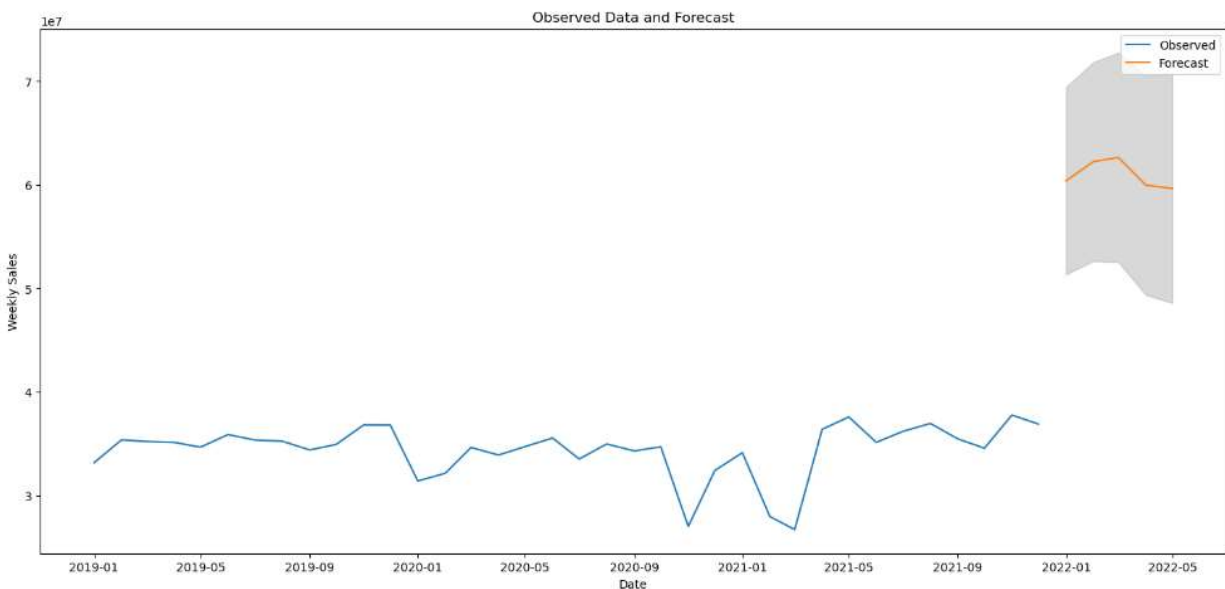


Fig 24: ARIMA Observed data and forecast

The plot provides a clear and concise visualization of how the forecasted values align with the observed data, while also conveying the level of uncertainty in the predictions through the confidence interval shading. This visualization helps stakeholders understand the accuracy and reliability of the forecasted sales values and make informed decisions based on this information

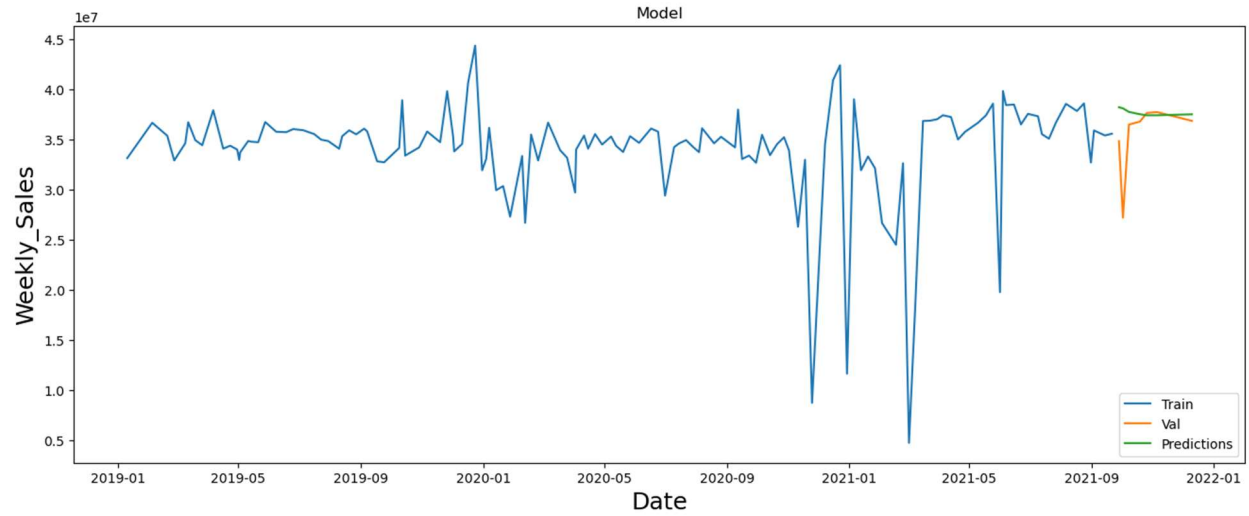


Fig 25: LSTM Forecasting plot

The plot provides a clear representation of the model's effectiveness in predicting weekly sales and allows stakeholders to make informed decisions based on the model's performance.



Fig 26: Forecasting plot for Attention Mechanism

This plot effectively communicates the performance of the attention mechanism model in predicting weekly sales, allowing stakeholders to evaluate its predictive accuracy and make informed decisions.

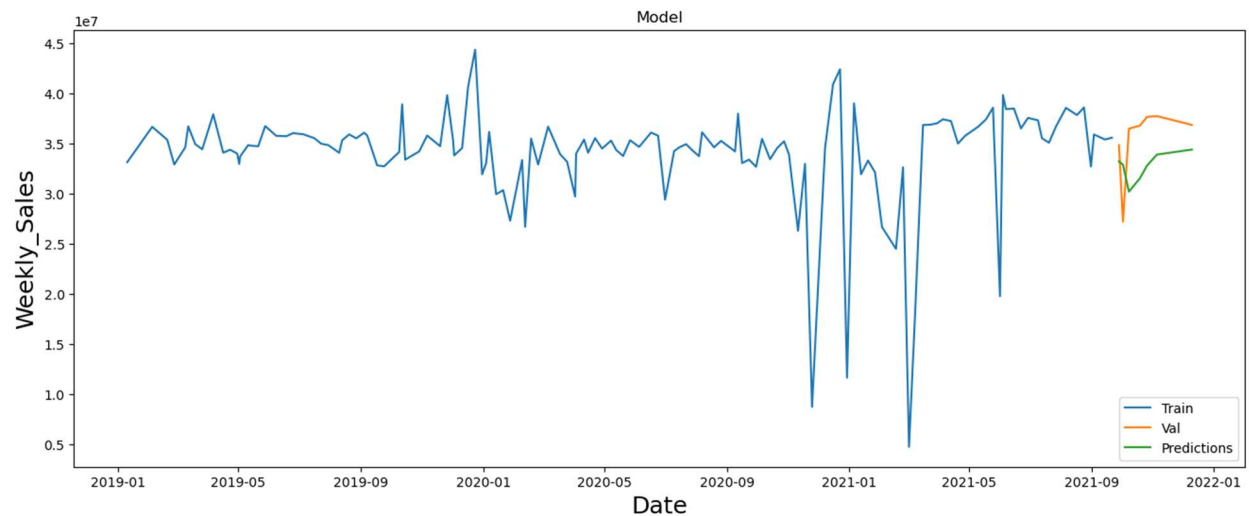


Fig 27: Forecasting plot for RNN

This plot succinctly presents the RNN model's performance in predicting weekly sales, aiding stakeholders in evaluating its efficacy and guiding decision-making.

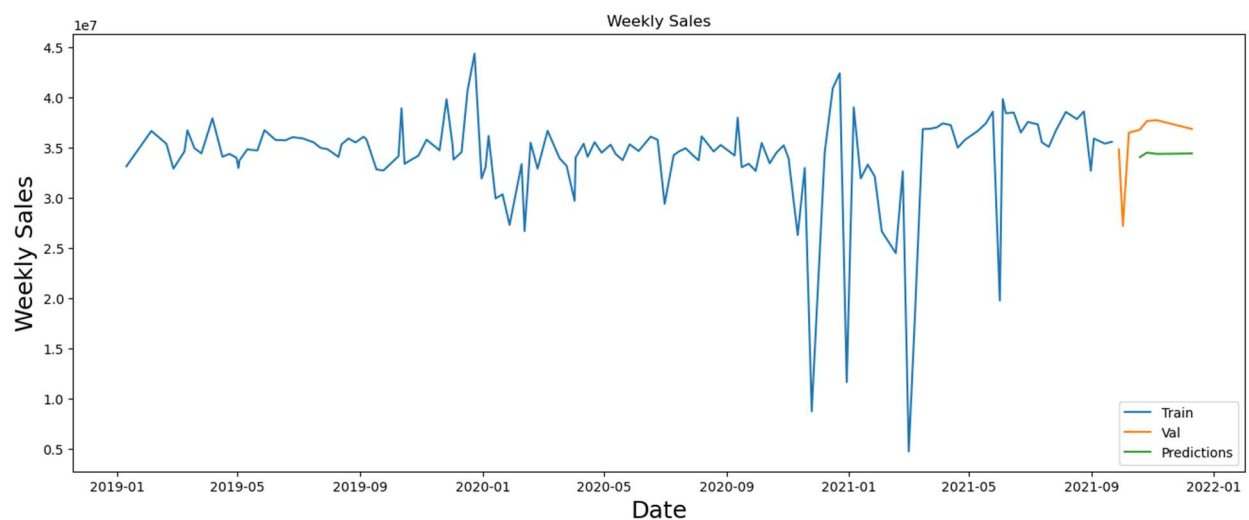


Fig 28: Forecasting plot for VAR

This plot effectively presents the performance of the VAR model in predicting Weekly Sales, aiding stakeholders in understanding its predictive capabilities and supporting decision-making processes.

Chapter 5

5. CONCLUSION AND FUTURE SCOPE

In the realm of predictive modeling, the thorough examination of model performance using metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) offers invaluable insights into the accuracy and reliability of diverse predictive models. These metrics play a pivotal role in assessing the models' capacity to generalize patterns and make predictions on new data. The analysis of different models based on these metrics not only aids in informed decision-making but also lays the groundwork for potential enhancements in the future.

The comprehensive analysis of various predictive models has unveiled distinctive performance trends. Mean Squared Error (MSE), serving as a foundational indicator, delineates the average squared deviation between predicted and actual values. A lower MSE signifies superior performance, denoting predictions that closely align with real-world values. Root Mean Squared Error (RMSE), an extension of MSE, furnishes an understandable scale congruent with the target variable's units. Conversely, Mean Absolute Percentage Error (MAPE) delves into the relative accuracy of forecasts, factoring in the percentage variance between predicted and actual values.

Among the evaluated models, the AutoRegressive Integrated Moving Average (ARIMA) model, while widely employed, grapples with complexities in capturing intricate temporal patterns. The comparatively elevated MSE, RMSE, and MAPE values highlight notable prediction disparities. In contrast, the Long Short-Term Memory (LSTM) model and Recurrent Neural Network (RNN) outshine ARIMA, demonstrating adeptness in capturing temporal intricacies and yielding diminished errors. The Graph Neural Network (GNN) and Vector Autoregression (VAR) models emerge as frontrunners, showcasing minimal errors and exemplifying proficiency in uncovering intricate data relationships.

The analysis reveals diverse patterns in how these techniques address the intricacies of sales forecasting. Attention mechanisms display a unique ability to weigh the importance of various input elements, while GNNs excel in capturing complex relationships among interconnected data points. RNNs, including LSTM, demonstrate their proficiency in harnessing temporal dependencies, which is pivotal for sales predictions. ARIMA and VAR, rooted in time series analysis, contribute their classical statistical strengths to the domain.

Future Scope: Looking ahead, the investigation of advanced machine learning techniques for sales forecasting suggests a horizon brimming with potential avenues for exploration. This study lays the groundwork for future endeavors in this domain. As a natural progression, researchers could delve into synergistic approaches by amalgamating attention mechanisms, GNNs, RNNs, LSTM, ARIMA, and VAR to extract a comprehensive predictive framework that harnesses their respective strengths. Moreover, refining these models for enhanced interpretability, especially attention mechanisms and GNNs, could bridge the gap between accurate predictions and actionable insights. Incorporating external variables such as economic indicators and promotional events into these techniques might amplify their predictive power. Addressing data scarcity through innovative augmentation strategies or adapting models to learn from sparse data holds promise for improving robustness. Additionally, enabling these models to swiftly adapt to real-time changes and adhering to ethical considerations in their deployment are pivotal paths for future exploration. This evolving landscape necessitates interdisciplinary collaboration, inviting expertise from various domains to unlock the full potential of advanced machine learning techniques in revolutionizing sales forecasting accuracy and applicability.

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